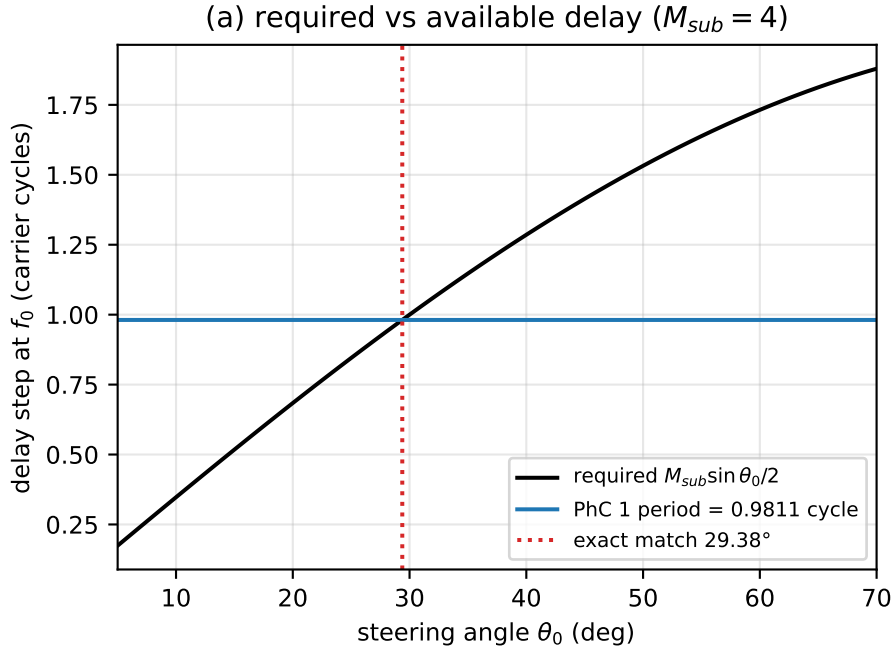
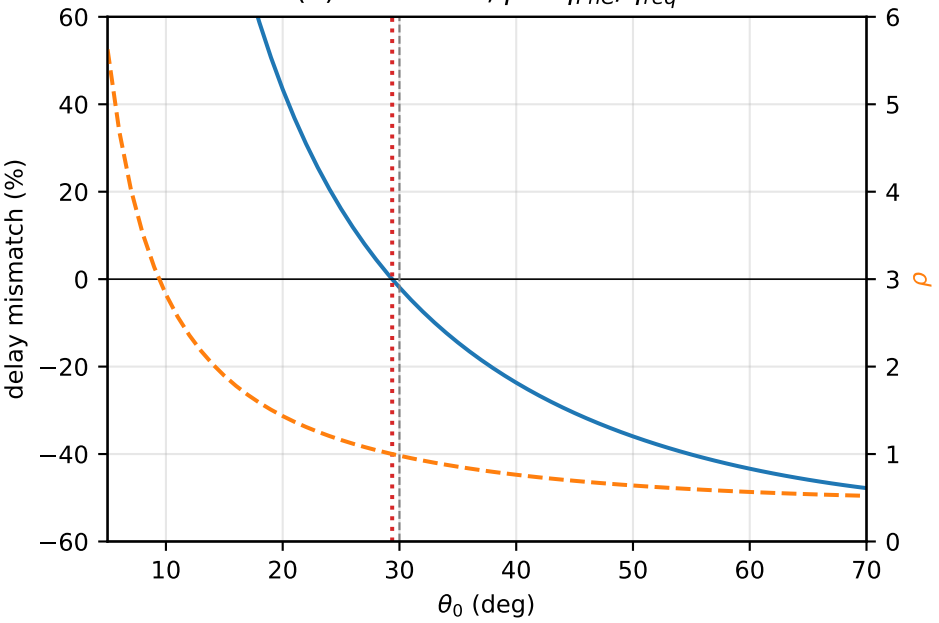
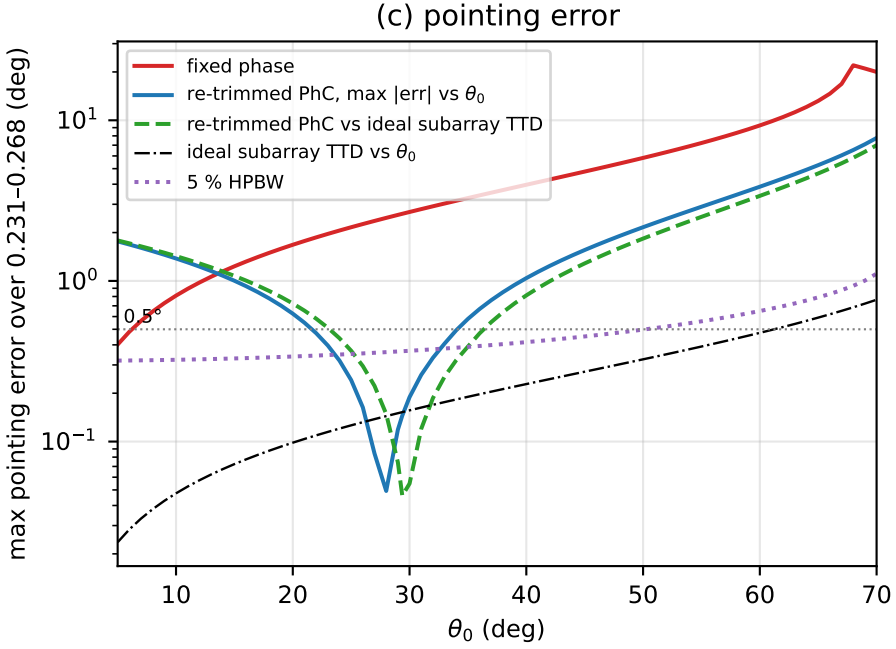
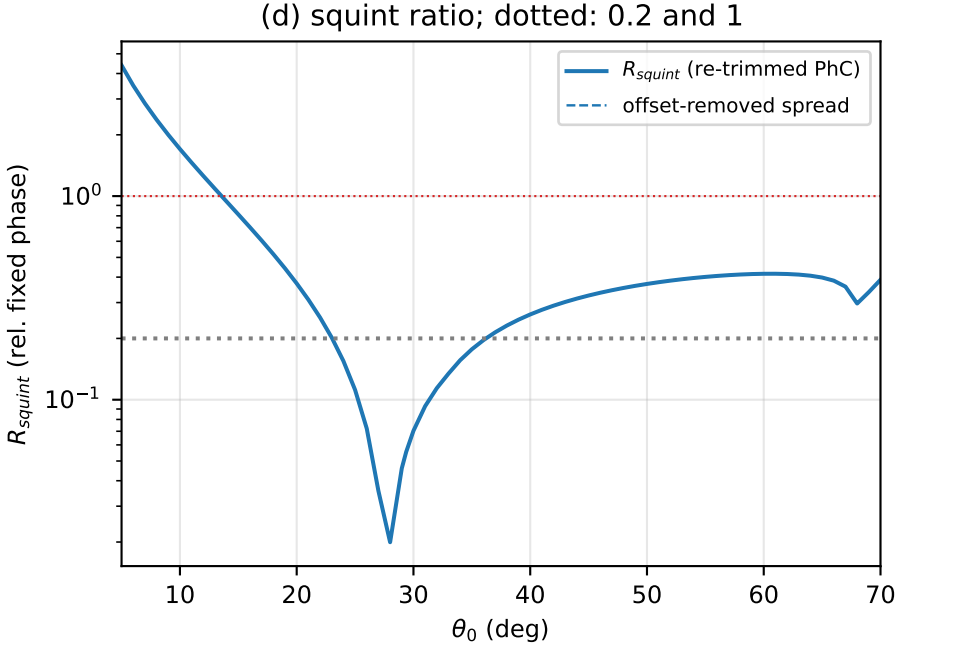

(a) required vs available delay ($M_{sub} = 4$)
delay step at f_0 (carrier cycles)
steering angle θ_0 (deg)
required $M_{sub}\sin\theta_0/2$
PhC 1 period = 0.9811 cycle
exact match 29.38°


(b) mismatch, $\rho = q_{PhC}/q_{req}$
delay mismatch (%)
ρ
θ_0 (deg)


(c) pointing error
max pointing error over 0.231–0.268 (deg)
fixed phase
re-trimmed PhC, max |err| vs θ_0
re-trimmed PhC vs ideal subarray TTD
ideal subarray TTD vs θ_0
5 % HPBW
0.5°
θ_0 (deg)


(d) squint ratio; dotted: 0.2 and 1
R_{squint} (rel. fixed phase)
R_{squint} (re-trimmed PhC)
offset-removed spread
θ_0 (deg)